OpenShift Interview Questions (Beginner to Advanced)

Beginner Level

1. What is OpenShift?

Answer: OpenShift is Red Hat's enterprise Kubernetes platform that helps deploy, manage, and scale containerized applications. It includes Kubernetes along with additional features like integrated CI/CD, built-in security, image registry, OperatorHub, monitoring, and a web console.

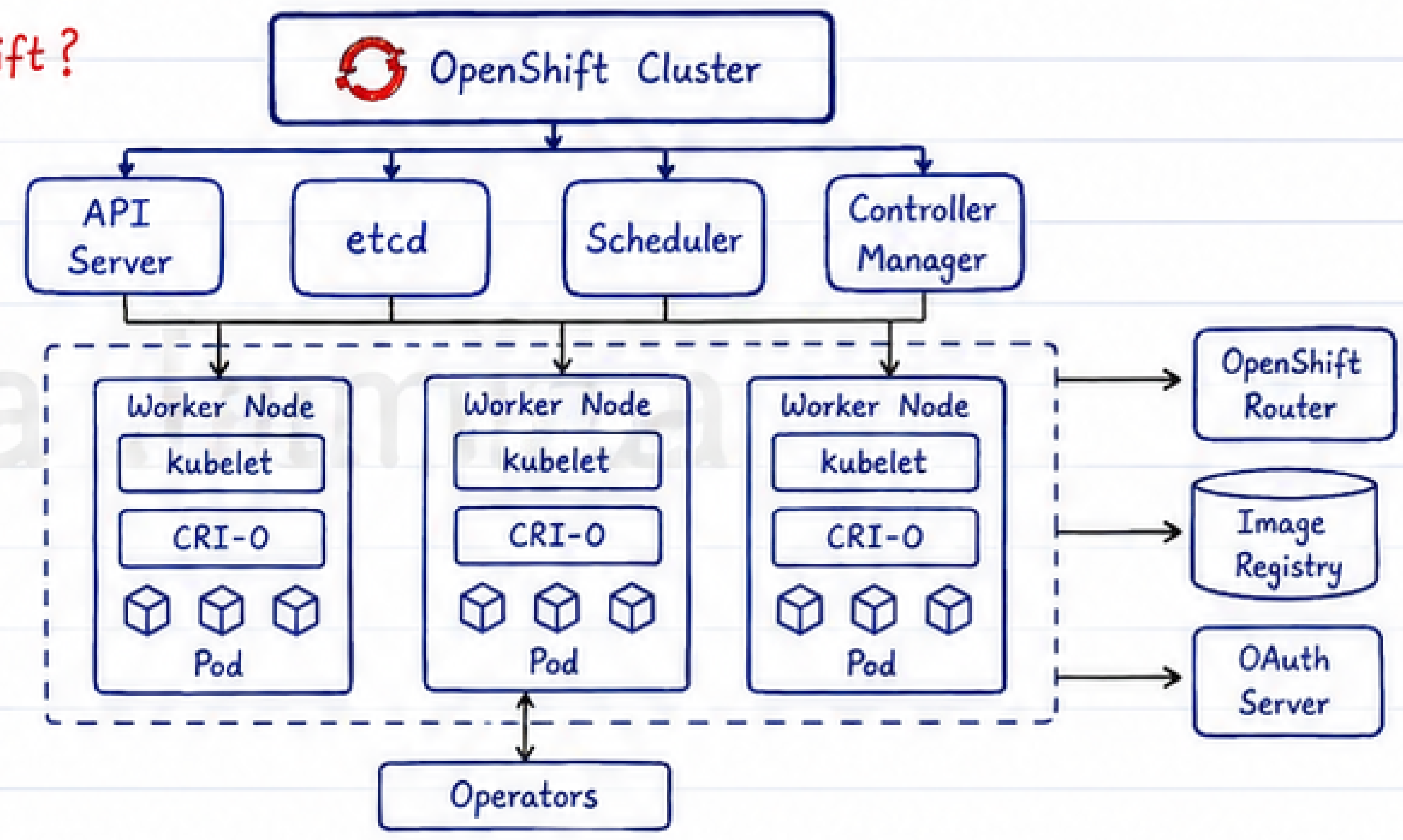
OPENSHIFT

2. What is the difference between Kubernetes and OpenShift?

Kubernetes	OpenShift
Open-source container orchestration platform	Enterprise Kubernetes platform by Red Hat
Manual installation of many components	Comes with integrated tools
Uses Ingress	Uses Routes
RBAC security	RBAC + Security Context Constraints (SCC)
No built-in image registry	Built-in Image Registry
Basic web dashboard	Full-featured Web Console

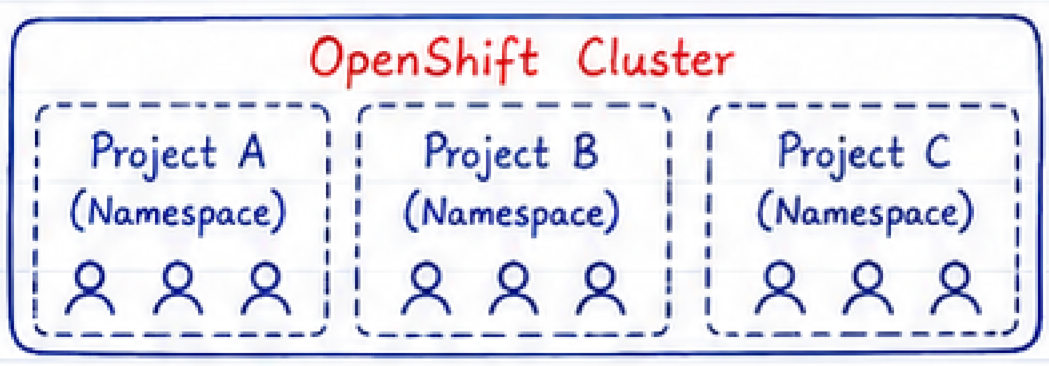
3. What are the main components of OpenShift?

- API Server
- etcd
- Scheduler
- Controller Manager
- Worker Nodes
- kubelet
- CRI-O
- OpenShift Router
- Image Registry
- Operators
- OAuth Server



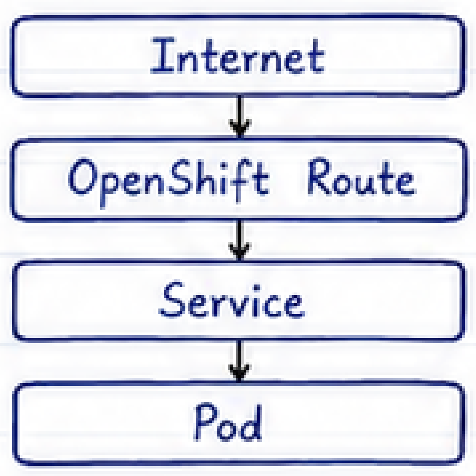
4. What is a Project in OpenShift?

Answer: A Project is OpenShift's implementation of a Kubernetes Namespace with additional user access controls and resource isolation.



5. What is an OpenShift Route?

Answer: A Route exposes an application running inside the OpenShift cluster to external users.



OpenShift Interview Questions (Beginner to Advanced)

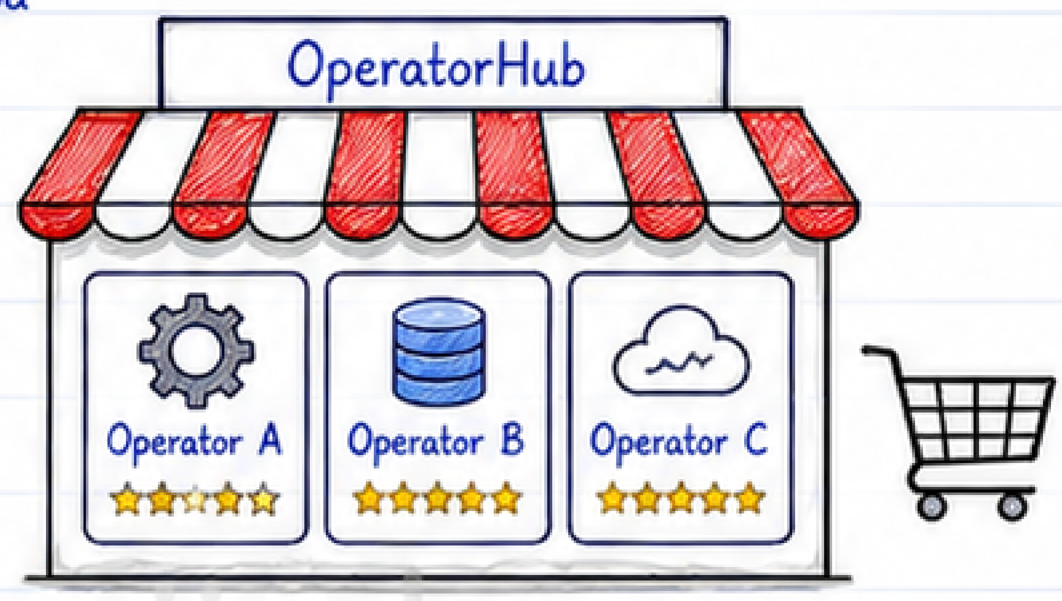
Beginner Level

6. What is SCC (Security Context Constraints)?

Answer: SCC controls the security permissions of Pods, such as whether containers can run as root, use privileged mode, or access host resources.

7. What is OperatorHub?

Answer: OperatorHub is the marketplace used to install and manage Operators in OpenShift.



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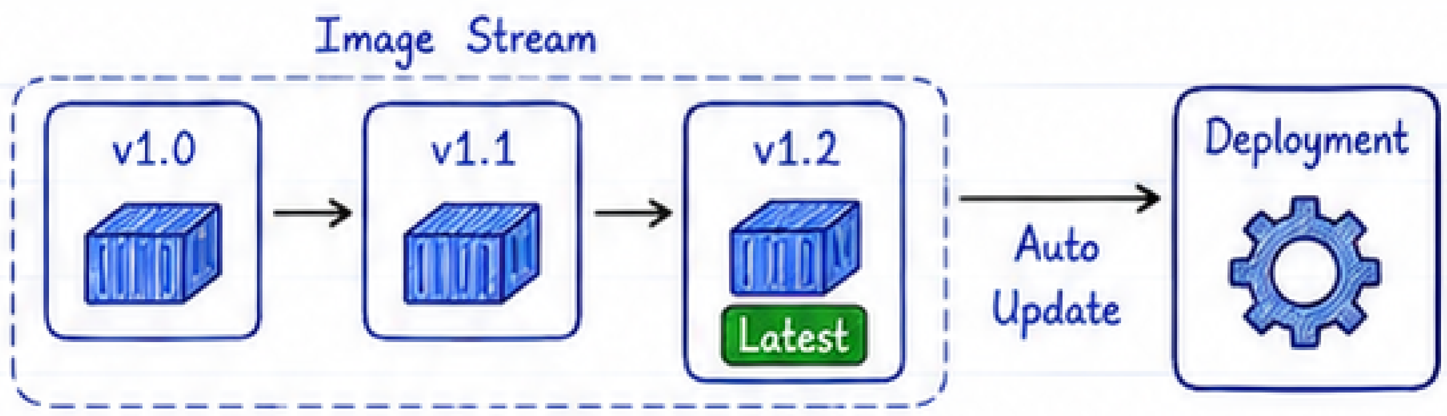
8. What are Operators?

Answer: Operators automate the installation, configuration, upgrades, scaling, monitoring, and recovery of applications running on OpenShift.

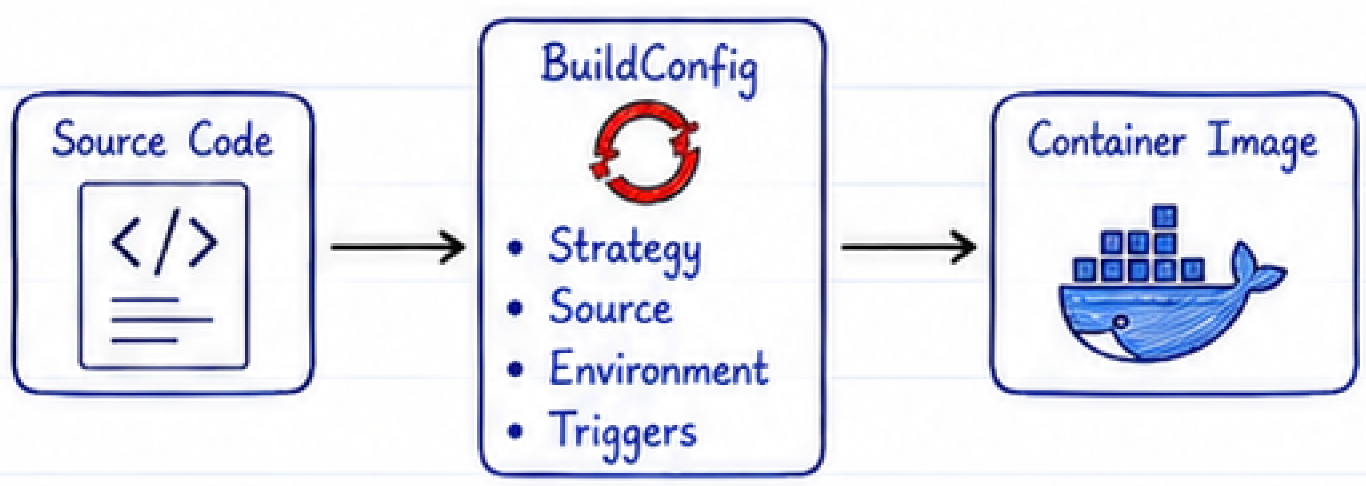
9. What is an Image Stream?

Answer: Image Streams track container images and automatically update deployments whenever a new image version becomes available.



10. What is BuildConfig?

Answer: BuildConfig defines how applications are built from source code or container images.



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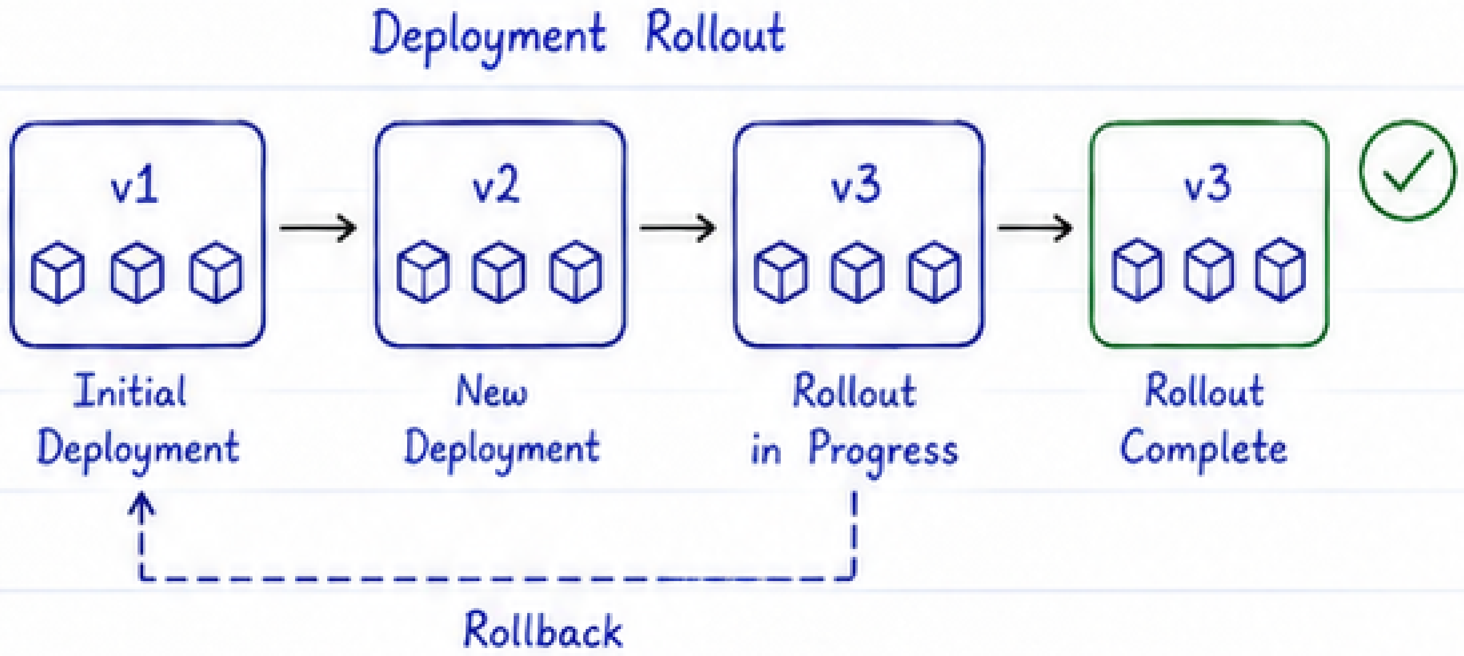
OpenShift Interview Questions (Beginner to Advanced)

Intermediate Level

11. What is DeploymentConfig?

Answer: DeploymentConfig is an OpenShift resource used to deploy and manage applications. It provides deployment triggers, rollbacks, and deployment strategies.



12. Deployment vs DeploymentConfig?

Deployment	DeploymentConfig
Kubernetes resource	OpenShift-specific resource
Managed by ReplicaSets	Managed by ReplicationControllers
Standard Kubernetes deployments	Supports Image Change Triggers and Config Change Triggers

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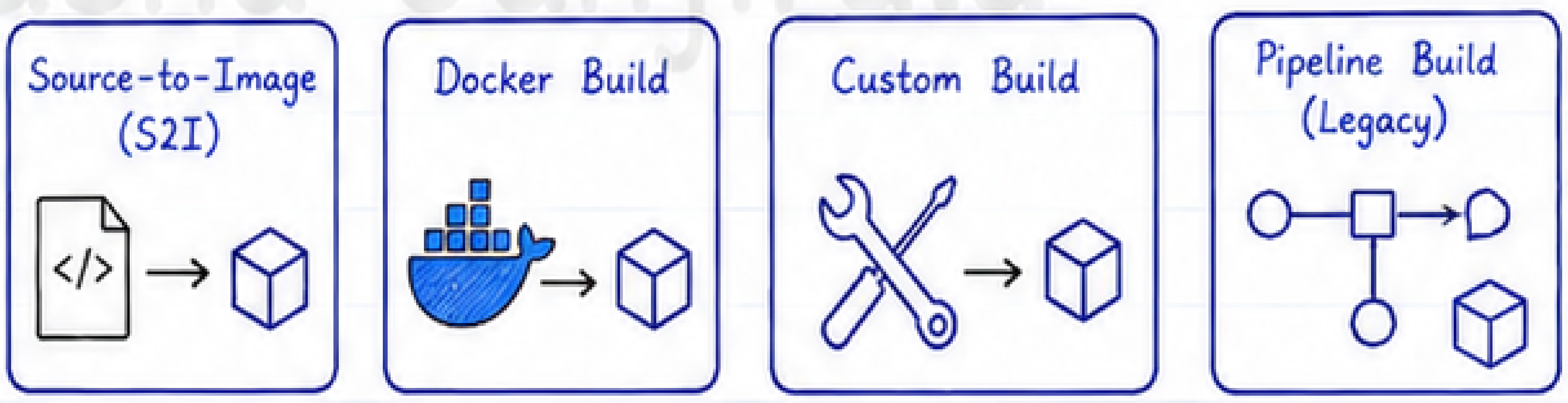
13. What are Build Strategies?

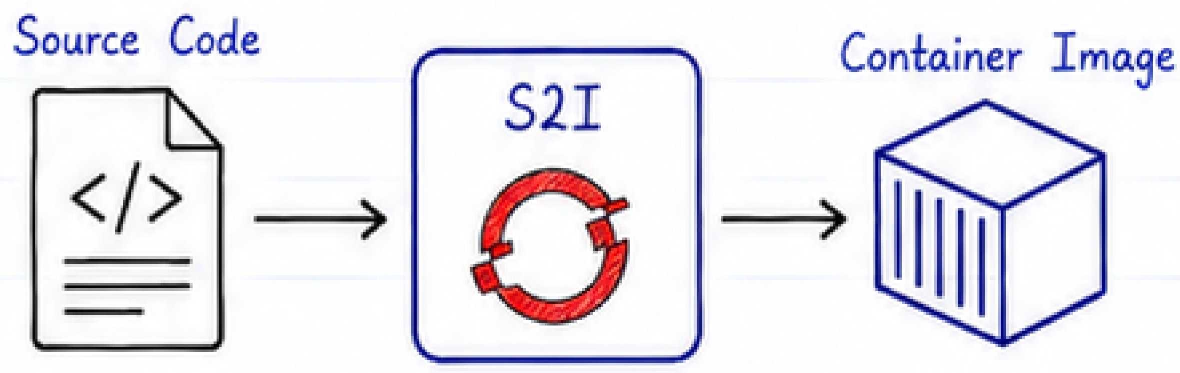
Answer: OpenShift supports:

- Source-to-Image (S2I)
- Docker Build
- Custom Build
- Pipeline Build (Legacy)



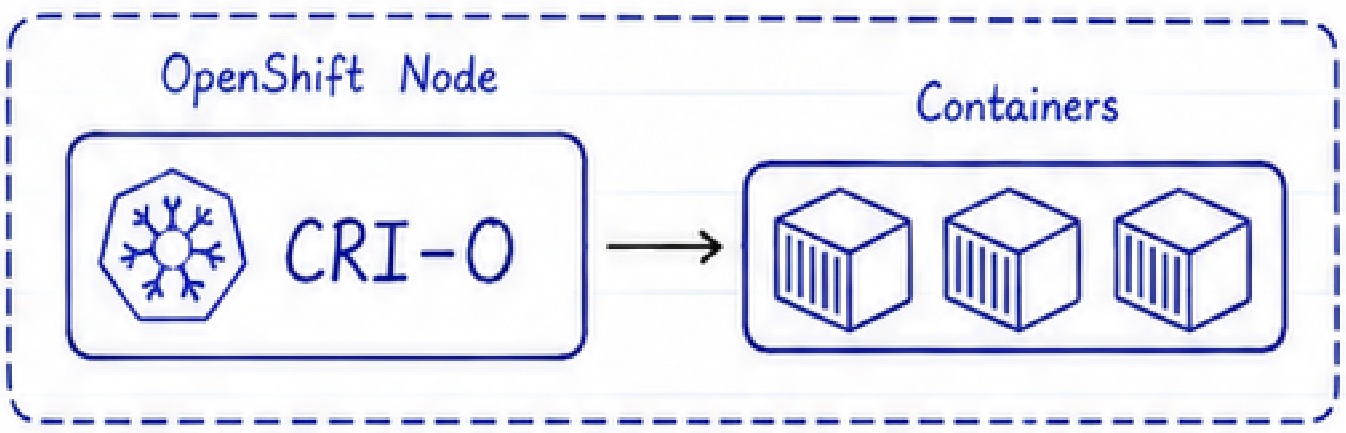
14. What is Source-to-Image (S2I)?

Answer: S2I automatically converts application source code into a runnable container image without requiring a Dockerfile.



15. What is CRI-O?

Answer: CRI-O is the container runtime used by OpenShift to run containers.

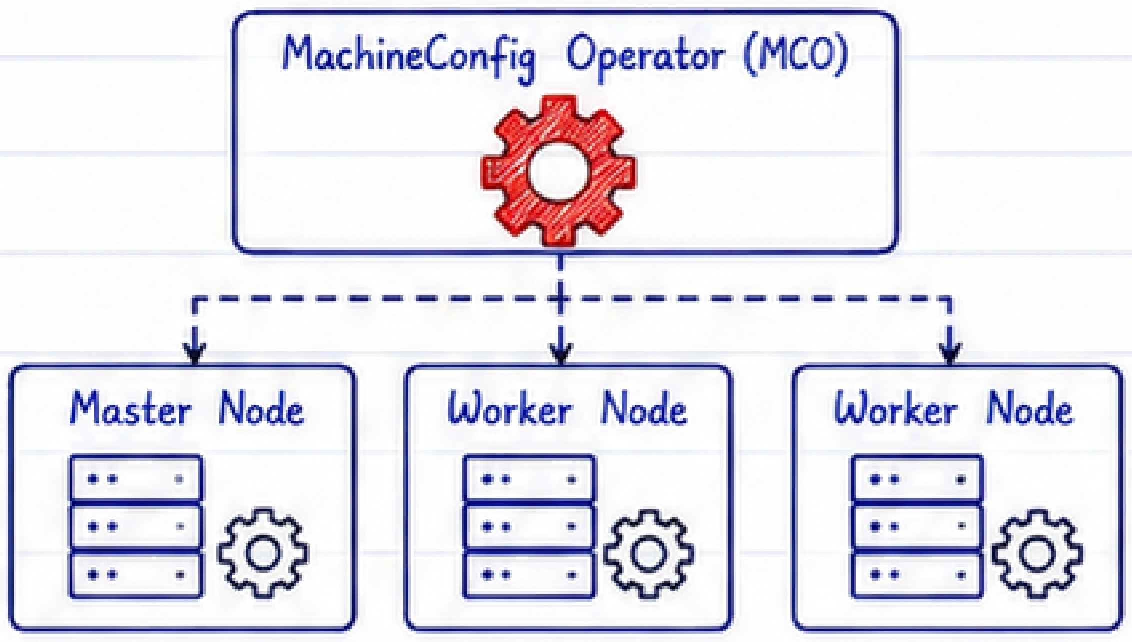


OpenShift Interview Questions (Beginner to Advanced)

Intermediate Level

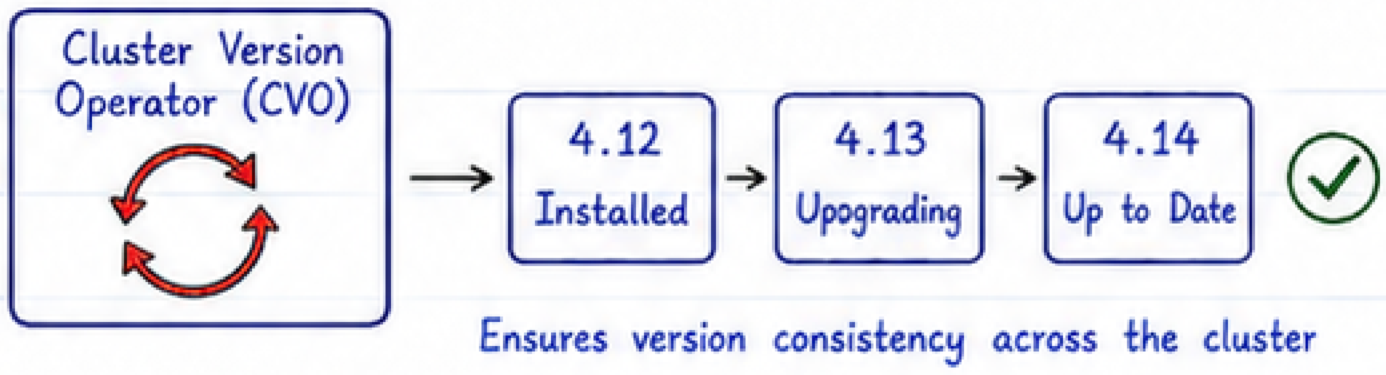
16. What is MachineConfig Operator (MCO)?

<u>Answer</u>: MCO manages operating system configuration across OpenShift worker and master nodes.



17. What is Cluster Version Operator (CVO)?

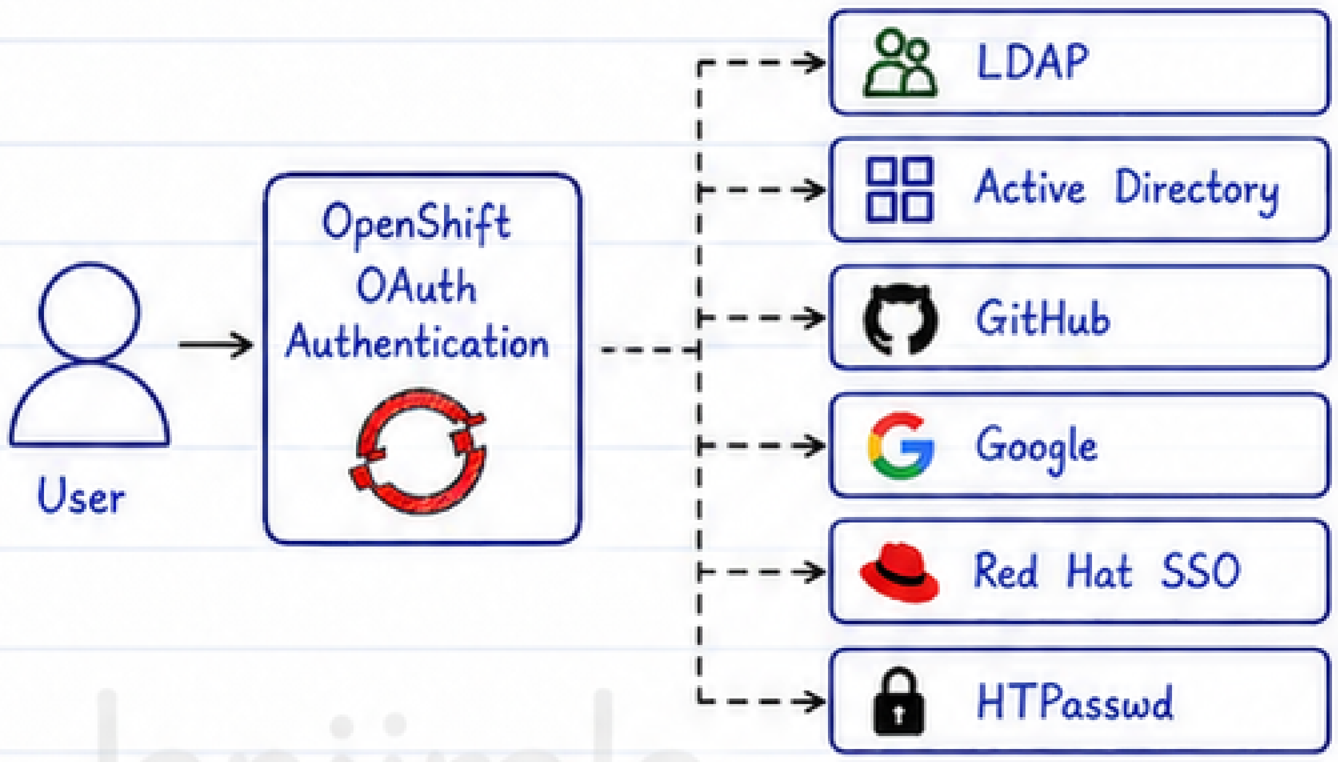
<u>Answer</u>: CVO manages OpenShift cluster installation, upgrades, and version consistency.



18. How does OpenShift authentication work?

<u>Answer</u>: OpenShift uses OAuth authentication and can integrate with:

- LDAP
- Active Directory
- GitHub
- Google
- Red Hat SSO
- HTPasswd



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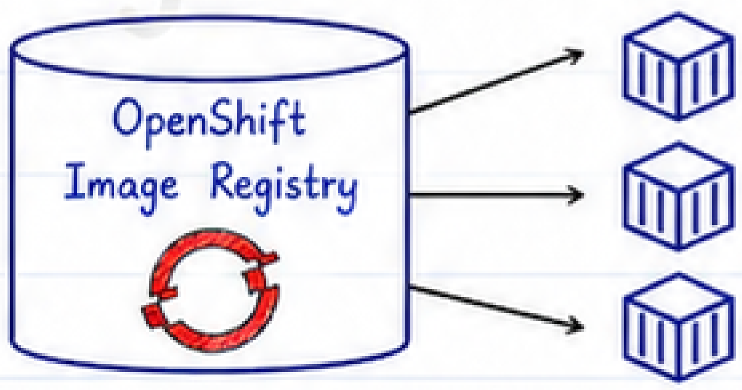
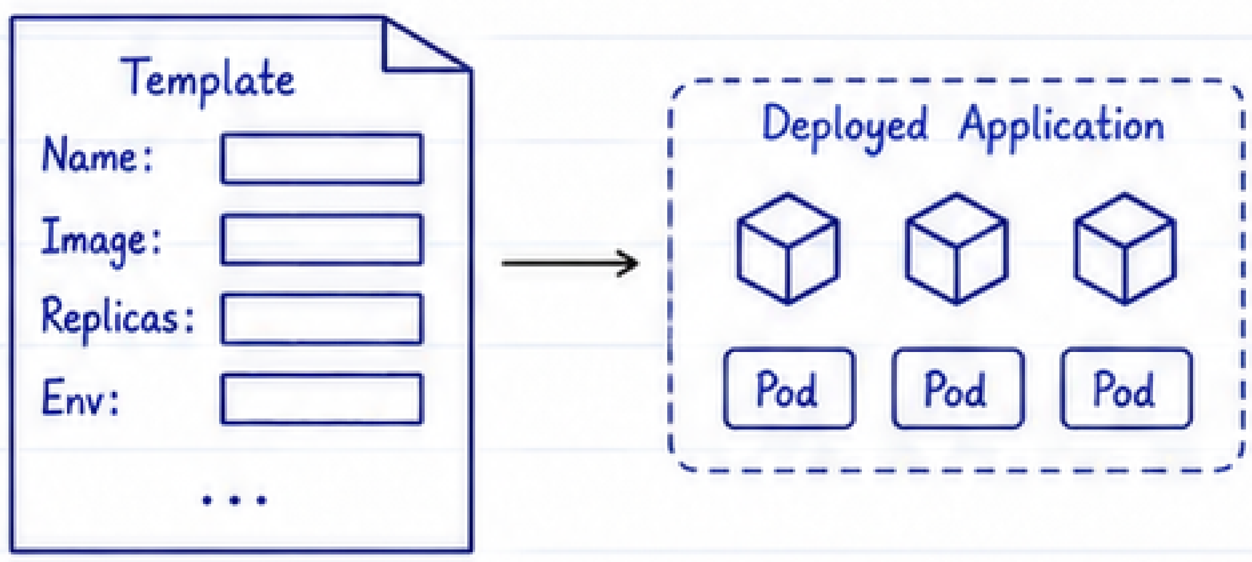
19. What is an Image Registry?

<u>Answer</u>: The integrated Image Registry stores container images used by OpenShift applications.



20. What are OpenShift Templates?

<u>Answer</u>: Templates allow reusable application deployments with configurable parameters.



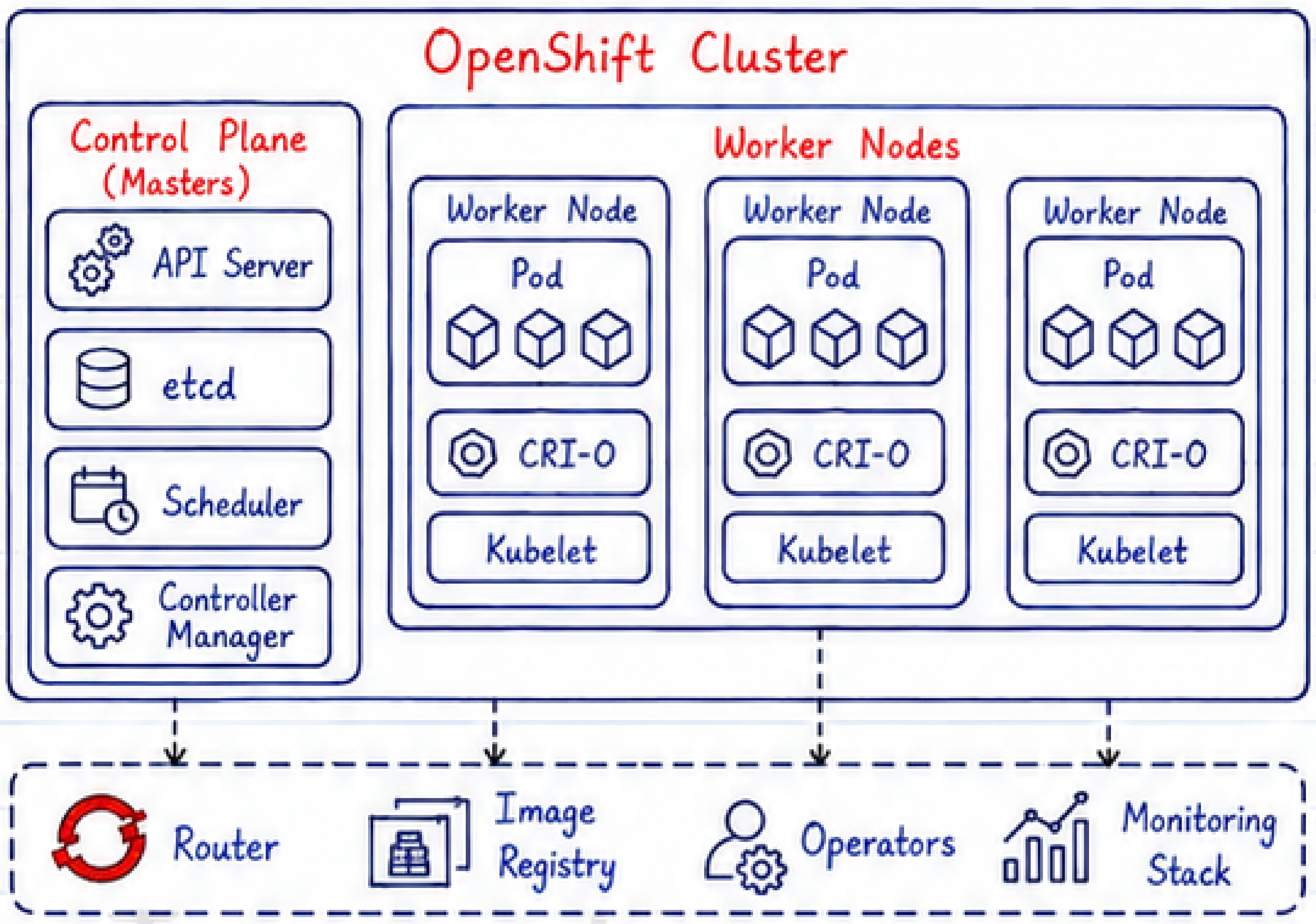
OpenShift Interview Questions (Beginner to Advanced)

Advanced Level

21. Explain the OpenShift Architecture.

Answer: OpenShift architecture consists of:

- Control Plane
- Worker Nodes
- API Server
- etcd
- Scheduler
- Controller Manager
- CRI-O
- Router
- Image Registry
- Operators
- Monitoring Stack



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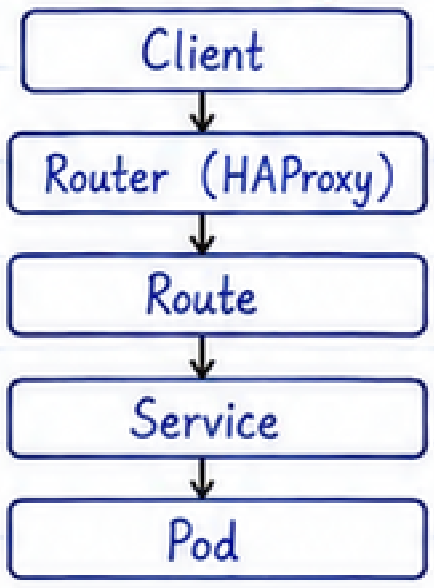
22. How does OpenShift Routing work?

Answer:



23. What is the OpenShift Monitoring Stack?

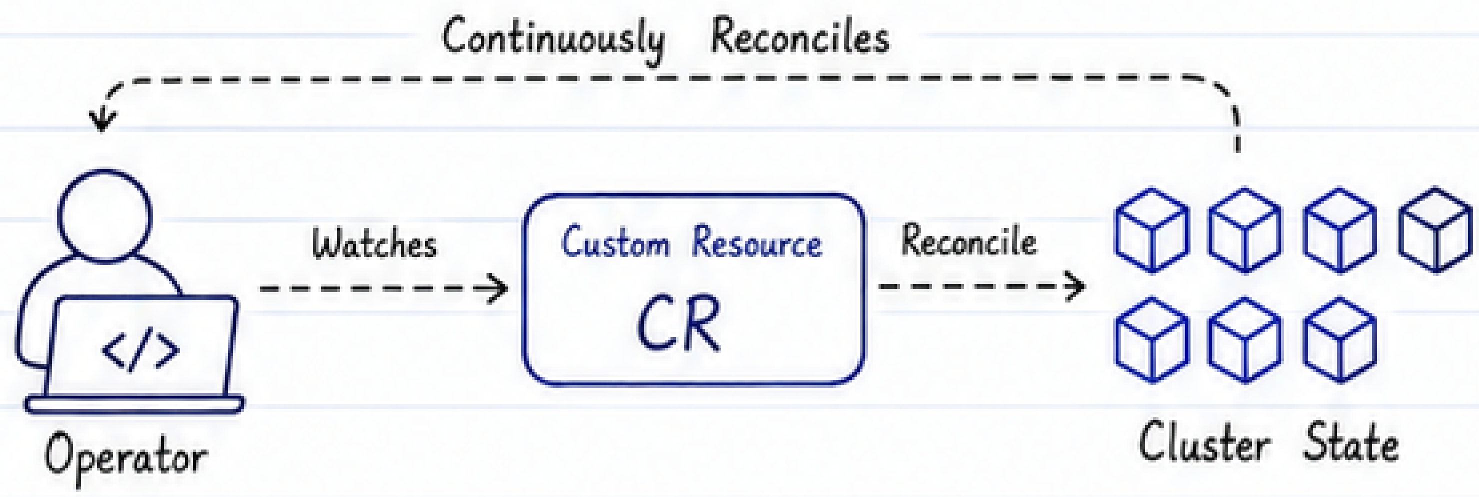
Answer: OpenShift includes:

- Prometheus
- Alertmanager
- Grafana
- Thanos

Prometheus Alertmanager Grafana Thanos

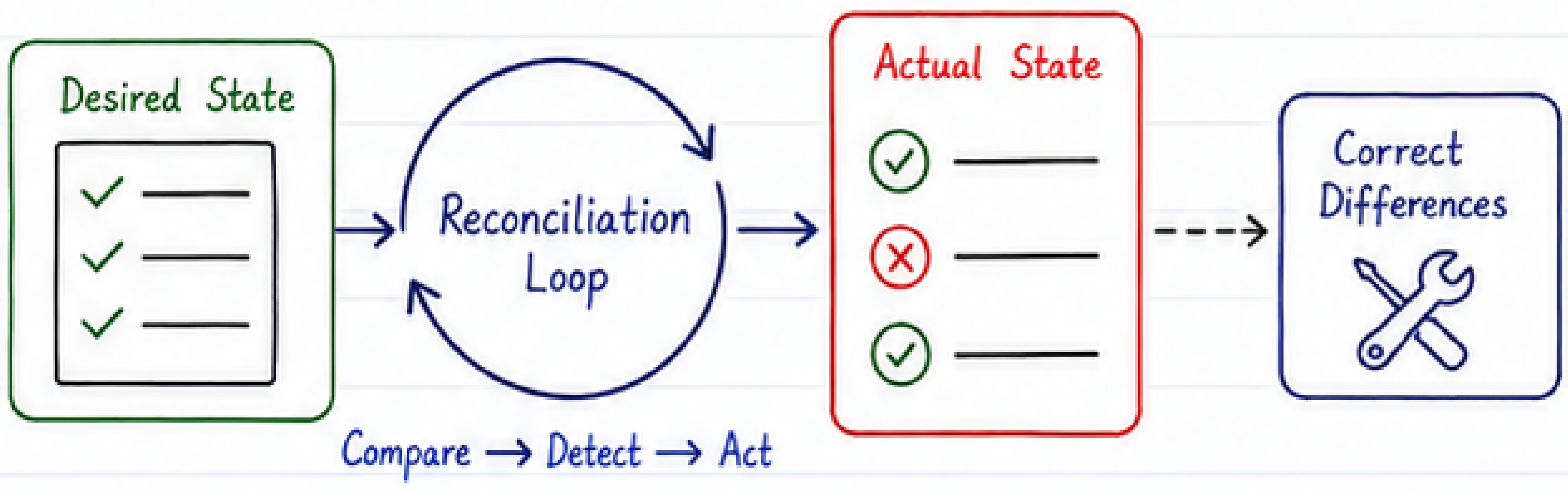
24. How do Operators work?

Answer: Operators watch Custom Resources and continuously reconcile the cluster to match the desired state.



25. What is the Reconciliation Loop?

Answer: It is the continuous process where Operators compare the desired state with the actual state and automatically correct any differences.

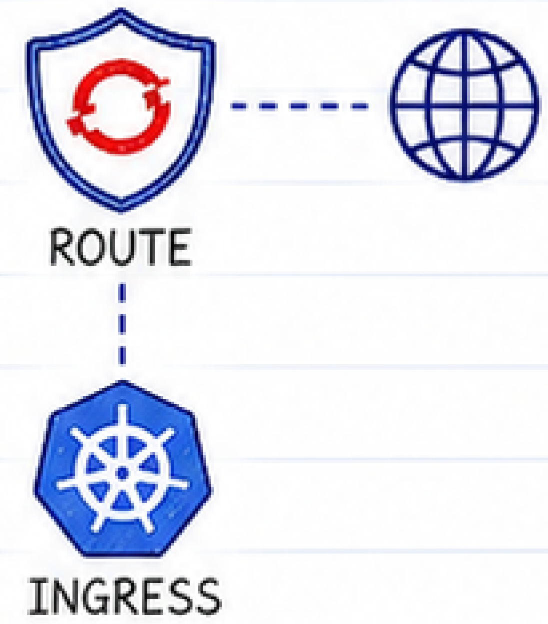


OpenShift Interview Questions (Beginner to Advanced)

Advanced Level

26. What is the difference between Routes and Ingress?

Route	Ingress
OpenShift-specific	Kubernetes standard
Managed by OpenShift Router	Managed by Ingress Controller
Supports edge, passthrough, and re-encrypt TLS	Depends on Ingress Controller features



27. How do you secure an OpenShift cluster?

Answer:

- Use RBAC
- Configure SCC
- Apply Network Policies
- Encrypt Secrets
- Enable TLS
- Scan container images
- Use least-privilege access
- Keep the cluster updated

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28. How do you upgrade OpenShift?

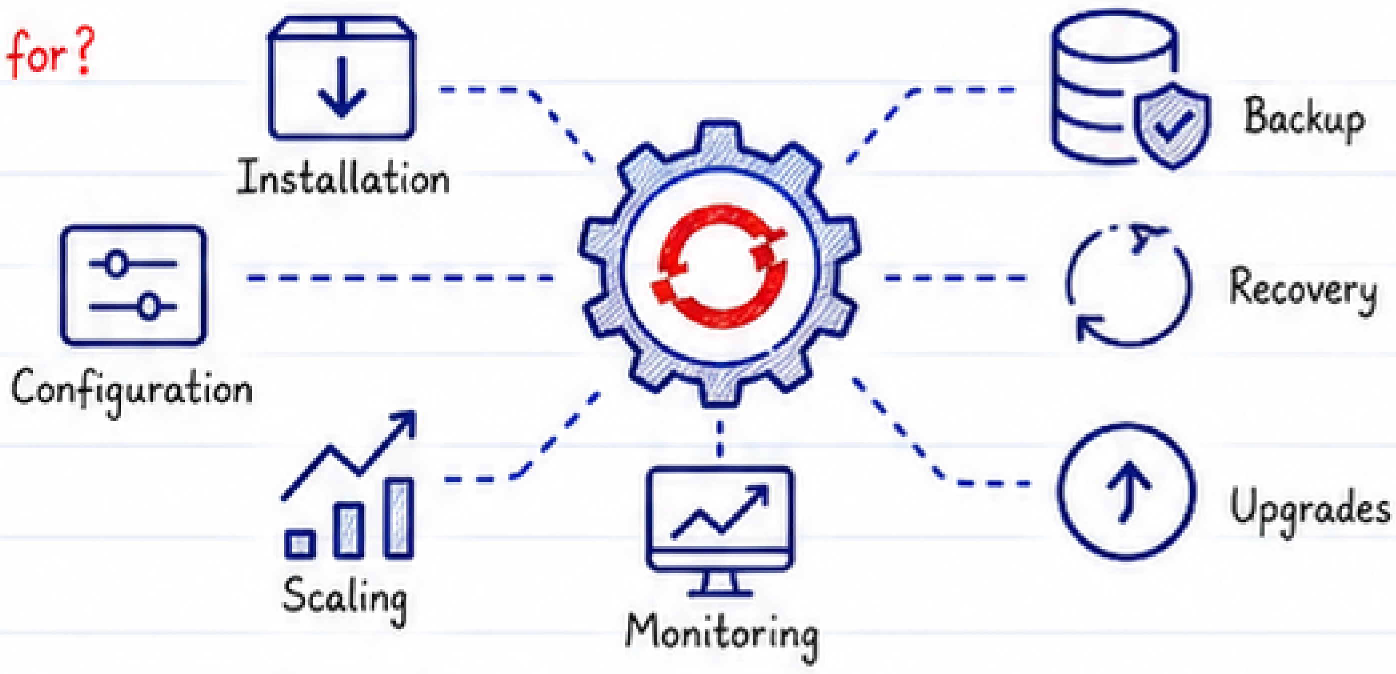
Answer:

OpenShift upgrades are managed automatically by the Cluster Version Operator (CVO), which coordinates the upgrade of cluster components and Operators.

29. What are OpenShift Operators used for?

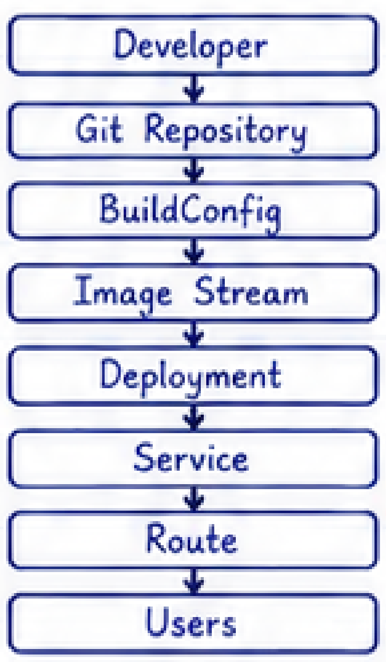
Answer: They automate:

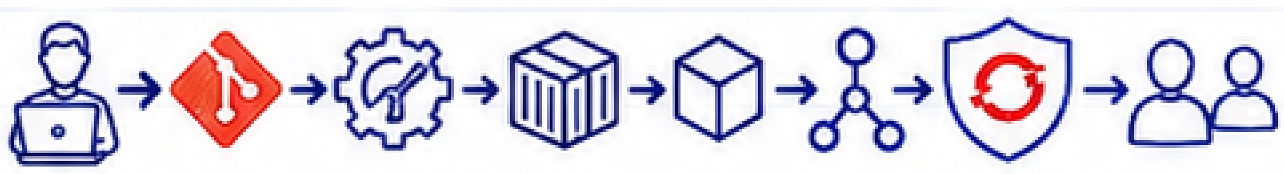
- Installation
- Configuration
- Scaling
- Monitoring
- Backup
- Recovery
- Upgrades



30. Explain the OpenShift CI/CD workflow.

Answer:





OpenShift Interview Questions (Beginner to Advanced)

Scenario-Based Questions

31. A Pod is stuck in Pending state. What will you check?

Answer:

- ☑ Node resources
- ☑ Scheduler events
- ☑ PVC availability
- ☑ Resource quotas
- ☑ Taints and tolerations
- ☑ Node selectors
- ☑ Events using `oc describe pod`

```
oc describe pod
Events:
Warning   FailedScheduling
0/3 nodes are available:
Insufficient cpu.
...
```

32. Users cannot access the application. What will you troubleshoot?

Answer:

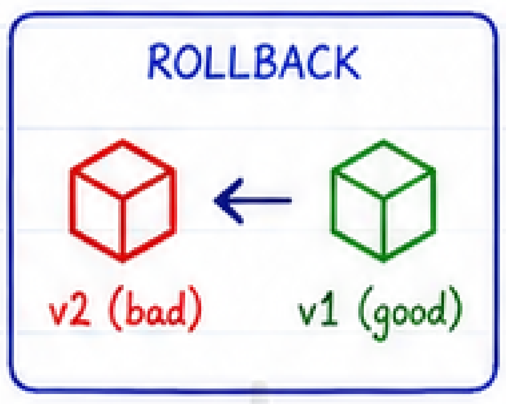
Check:

- ☑ Pod status
- ☑ Service configuration
- ☑ Route
- ☑ Router logs
- ☑ DNS resolution
- ☑ TLS certificates
- ☑ Network Policies

33. A deployment failed after a new image was pushed. What would you do?

Answer:

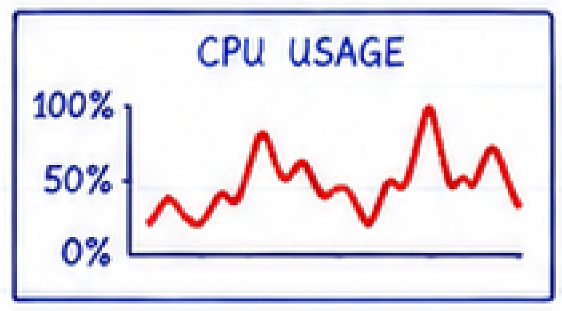
- ☑ Check ImageStream
- ☑ Verify image availability
- ☑ Review deployment logs
- ☑ Roll back to the previous image if necessary
- ☑ Validate BuildConfig and DeploymentConfig



34. How do you investigate high CPU usage in OpenShift?

Answer:

- ☑ Use `oc adm top nodes`
- ☑ Use `oc adm top pods`
- ☑ Check Prometheus metrics
- ☑ Review Grafana dashboards
- ☑ Analyze application logs
- ☑ Scale the application if needed



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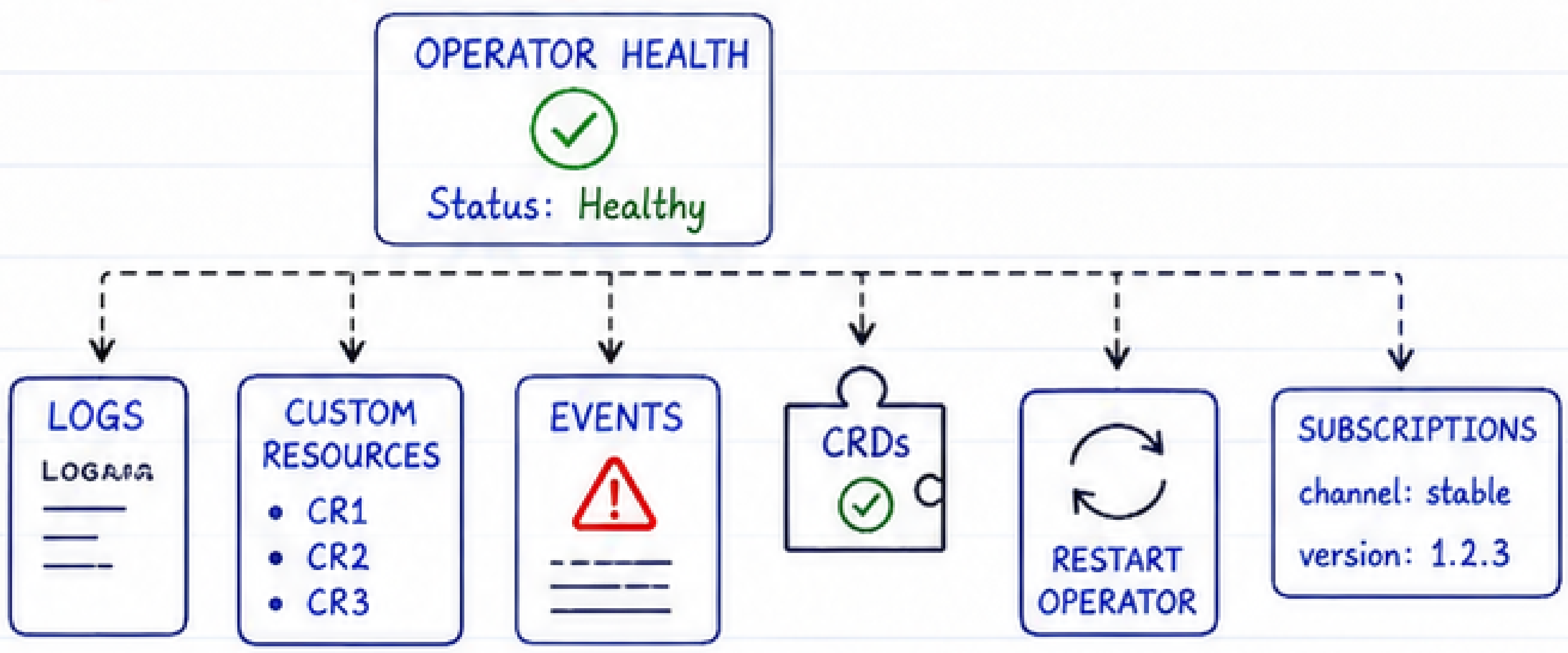
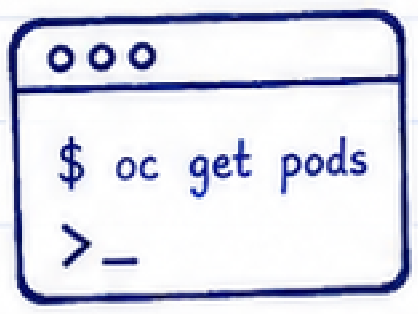
35. An Operator is not functioning correctly. What steps would you take?

Answer:

- ☑ Check Operator status
- ☑ Review Operator logs
- ☑ Verify Custom Resources
- ☑ Inspect Events
- ☑ Confirm CRDs exist
- ☑ Restart the Operator if necessary
- ☑ Check Operator subscriptions and versions



OpenShift Interview Questions (Beginner to Advanced)



Frequently Asked Commands

Command	Purpose
oc login	Log in to an OpenShift cluster
oc project	Switch projects
oc get pods	List Pods
oc get nodes	List cluster nodes
oc get routes	View Routes
oc get svc	List Services
oc get deployments	List Deployments
oc describe pod <pod>	View Pod details
oc logs <pod>	View Pod logs
oc exec -it <pod> -- bash	Access a running container
oc apply -f file.yaml	Apply a resource configuration
oc delete -f file.yaml	Delete a resource
oc rollout status deployment/<name>	Check deployment status
oc rollout restart deployment/<name>	Restart a deployment
oc adm top nodes	View node resource usage
oc adm top pods	View pod resource usage

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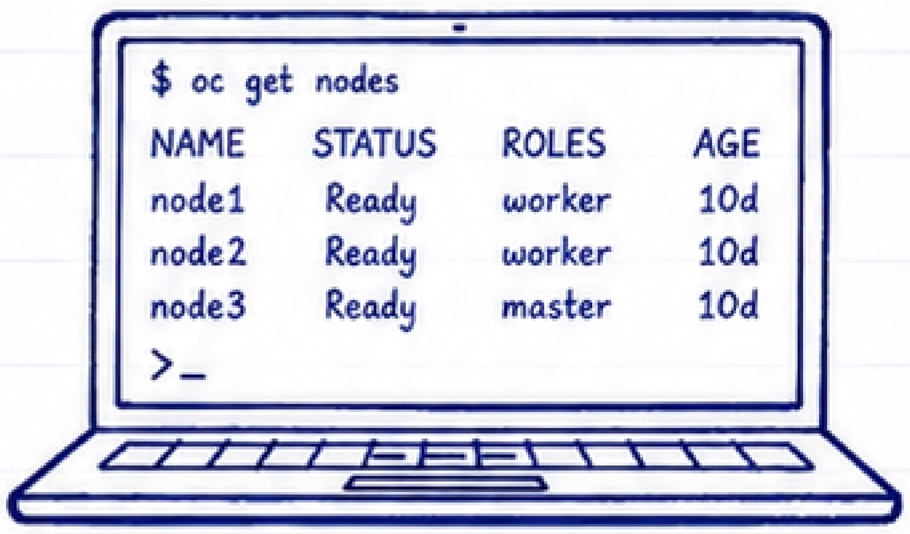
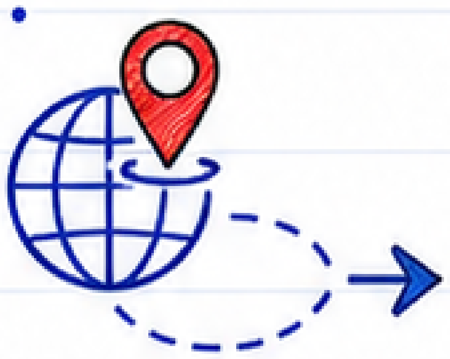
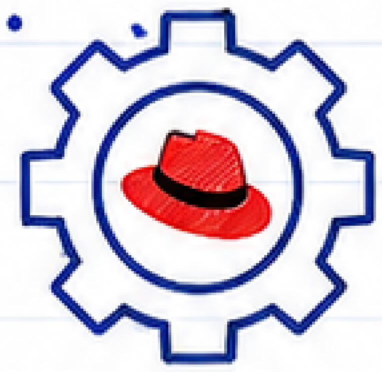
```
$ oc get pods
NAME                     READY   STATUS    RESTARTS   AGE
web-7f5d6c7f6f-abc12     1/1     Running   0          2m
>_
```



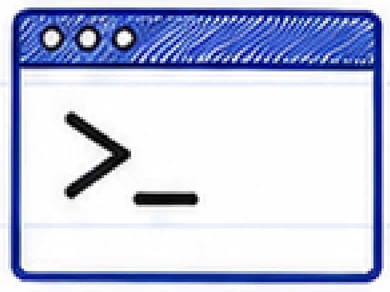
Quick Revision Table

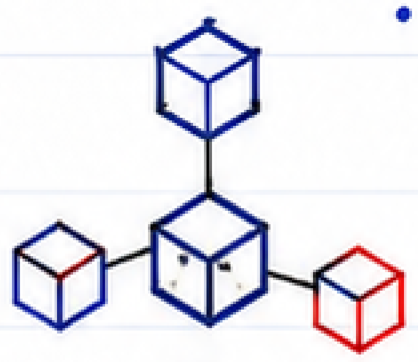
OPENSHIFT





Topic	One-Line Summary
OpenShift	Enterprise Kubernetes platform by Red Hat.
Project	OpenShift's enhanced version of a Kubernetes Namespace.
Route	Exposes applications externally.
SCC	Controls Pod security permissions.
Operator	Automates application lifecycle management.
Image Stream	Tracks and manages container image versions.
BuildConfig	Defines how applications are built.
DeploymentConfig	OpenShift deployment resource with triggers.
CRI-O	Container runtime used by OpenShift.
Monitoring Stack	Includes Prometheus, Alertmanager, Grafana, and Thanos.
oc CLI	Command-line tool for managing OpenShift clusters.
Troubleshooting	Verify Pods, Routes, Services, Events, Logs, and Resources.





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